

Abdulmajid Awol Seid

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Summary

PhD student in Computer Science at Tulane University and graduate research assistant in the Laboratory for Software Design, advised by Dr. Hridesh Rajan. I research reliable and secure AI agents at the intersection of artificial intelligence, software engineering, and programming languages. My current work focuses on the reliability and security of tool-using AI agents.

Research Interests

Reliable AI agents: Agentic systems, tool use, failure analysis, evaluation

Agent safety and security: Tool misuse, untrusted inputs, provenance

AI + software engineering: AI4SE, agentic software engineering, empirical software engineering

PL for AI agents: Contracts, state, refinement, provenance, specification

Education

PhD	Tulane University , Computer Science - School of Science and Engineering. Advisor: Dr. Hridesh Rajan. - Research focus: AI/LLM reliability, AI agent safety, and software engineering.	New Orleans, LA, USA Aug 2026 – present
MSc	Addis Ababa Science and Technology University , Software Engineering (Fast Track Program) - Thesis: <i>Predicting High-Risk Technical Debt in Open-Source Software Projects Using Machine Learning</i>. Advisor: Dr. Tesfaye Gidey. - Scholarship for academic excellence to begin the master's before completing the bachelor's degree.	Addis Ababa, Ethiopia Jan 2024 – July 2026
BSc	Addis Ababa Science and Technology University , Software Engineering Graduated with Honors (Very Great Distinction), GPA 3.78/4.0.	Addis Ababa, Ethiopia May 2021 – June 2025
	Africa to Silicon Valley (A2SV) , Software Engineering and Algorithms Training 1,000+ hours of data structures and algorithms training; 600+ problems solved on LeetCode and Codeforces.	Addis Ababa, Ethiopia Nov 2023 – Nov 2024

Thesis

Predicting High-Risk Technical Debt in Open-Source Software Projects Using Machine Learning	July 2026
Abdulmajid Awol Seid github.com/maajidAwol/technical-debt (Master's thesis, Addis Ababa Science and Technology University)	

Research Experience

Tulane University, Laboratory for Software Design , Graduate Research Assistant - Researching the reliability and security of tool-using AI agents. - Exploring software engineering and programming-language techniques (contracts, state, provenance) for reasoning about agent behavior.	New Orleans, LA, USA Aug 2026 – present
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Addis Ababa Science and Technology University, MSc Thesis Research

- Built a reproducible cross-project framework that predicts which files become maintenance-intensive (27 features, 22 Apache Java projects). On unseen projects, the top 20% of files ranked by the best model recover about 82% of future maintenance-intensive files, about 4.1x random inspection.

Addis Ababa, Ethiopia
Completed July 2026

Engineering Experience

Ethiopian Artificial Intelligence Institute, Software Engineer

Addis Ababa, Ethiopia
Aug 2025 – June 2026

Upwork (Freelance), including Omnicasa, Brussels, Full Stack Developer

Remote

- Built a distributed video processing pipeline with Redis-backed job queues and asynchronous workflows, cutting turnaround from 2 hours to 15 minutes per recording.
- Automated transcription and post-meeting analytics, reducing review time by 30%. Top Rated freelancer.

July 2024 – Mar 2026

iCog Labs, AI/ML Intern

Addis Ababa, Ethiopia
May 2025 – Aug 2025

AASTU Software Engineers Association, Software Engineer Associate

Addis Ababa, Ethiopia
Sept 2024 – June 2025

Eskalate LLC, Mobile Application Developer

- Built Flutter mobile applications in a team following test-driven development.

Addis Ababa, Ethiopia
June 2024 – Nov 2024

Camara Education Ethiopia, Full Stack Developer Intern

- Contributed to an educational platform used by 200+ Ethiopian schools and 300,000+ students.

Addis Ababa, Ethiopia
July 2023 – Nov 2023

Selected Projects

Miiltoo Knowledge Companion

Aug 2023 – Nov 2023

- Led a team of five building a generative-AI chatbot and quiz bot for Ethiopian Grade 12 exam preparation (Flask, MySQL, OpenAI API).
- Top 9 of 3,700+ participants from 47 countries, A2SV Generative AI for Africa Hackathon.

Flutter Clean Architecture VS Code Extension

Dec 2024 – Mar 2025

- Open-source extension (190+ installs) that scaffolds and auto-repairs Clean Architecture structure in Flutter apps.

Honors and Awards

2025: Academic Excellence Award, Graduated with Honors (Very Great Distinction), AASTU

2024: MSc Fast Track Scholarship for Academic Excellence, AASTU

2023: Finalist (Top 9 of 3,700+), A2SV Generative AI Hackathon for Africa

2021: University Presidential Award, AASTU

Skills

Programming: Python, Java, C++, JavaScript, TypeScript, Dart, PHP, SQL

AI / ML: Supervised learning (logistic regression, random forests, XGBoost, LightGBM), model evaluation, LLM APIs, RAG

Software engineering: Software architecture, testing, distributed and asynchronous systems, CI/CD, Git, Docker

Research methods: Empirical software engineering, ML for software engineering, cross-project validation, reproducible pipelines